

Comment on “Descriptor: Heart and Lung Sounds Dataset Recorded from a Clinical Manikin using Digital Stethoscope (HLS-CMDS)”

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September 22, 2026

Status: DRAFT, ready for review. Both open items are now closed. Provenance: the copy audited here is byte-identical to the Mendeley release — all six top-level artifacts match on MD5, verified 2026-08-19 — so the findings concern the released dataset rather than a redistribution. Sample rate: checked against the descriptor’s own text (Table I, p. 138, and the comparative claim on p. 134), which changed how Section 4 is phrased — 22 050 Hz is stated there as an instrument specification, and the descriptor never states the released files’ rate at all.

Abstract

While using the HLS-CMDS dataset [1] to build an evaluation pipeline for heart/lung sound source separation, we ran independent integrity checks against the paired `Mix.csv` recordings that `Mix.csv` claims to be additive combinations of the isolated `HS.csv`/`LS.csv` recordings. Two independent methods — a least-squares additivity fit and a clipping-artifact scan — agree that only 36 of the 145 published mixture rows are acoustically consistent with their stated heart/lung sources; the remaining 109 are not. We also report a sample-rate gap: every released file measures 4,000 Hz, while the only sample rate the descriptor gives is 22,050 Hz, stated as a specification of the recording instrument; the section describing the released files does not state their rate at all. We report both findings here, separately from our own separation-methods work, because the first bears on ground truth already in use downstream. Of the four published studies we are aware of that build on HLS-CMDS, two evaluate on all 145 rows, and for one of them the exposure is measurable from its own tables: 4,033 of the 5,365 scored windows behind its headline result fall on rows we find non-additive (Section 5).

1 Motivation for this comment

This comment is deliberately kept separate from our companion methods paper (in preparation) on leakage-safe heart/lung separation evaluation. That paper’s contribution is a separation pipeline and baseline comparison; the findings below are about the integrity of the released dataset itself,

and are relevant to every downstream user of HLS-CMDS, not only to our own results. We report them here, addressed to the descriptor’s own venue, so they are discoverable by anyone consulting the dataset in the future.

2 Additive-triplet audit

HLS-CMDS’s `Mix.csv` lists, for each of 145 mixture recordings, the isolated heart (`HS.csv`) and lung (`LS.csv`) recording IDs that were combined to produce it. We tested this claim directly: for each row, we fit the least-squares scalar gain a minimizing $\|\text{mixed} - a \cdot (\text{heart} + \text{lung})\|$, then computed the relative residual $\|\text{mixed} - a \cdot (\text{heart} + \text{lung})\|/\|\text{mixed}\|$. A row is additive when this residual falls below 10^{-3} .

The result is sharply bimodal, not a continuum: genuinely additive rows land at relative residual $\sim 10^{-8}$ – 10^{-4} (consistent with 16-bit quantization noise on an exact sum), while the remaining rows land at relative residual ~ 1 — i.e., the mixed recording shares essentially no structure with its named heart/lung sources. Only **36 of 145 rows** (M0087, M0111–M0145) pass; the other 109 fail by a wide margin, not a borderline one.

This check is implemented as `verify_additive_triplets()` in our public repository (Section 7) so it is independently re-runnable against any copy of the dataset, rather than resting on a one-off finding.

3 Independent corroboration: a clipping fingerprint

A separate, unrelated audio-quality scan — flagging the fraction of samples pinned at the 16-bit full-scale rail (± 32767) in every `.wav` file referenced by `HS.csv/LS.csv/Mix.csv` — turned up a single-sample clipping artifact in exactly one subset of `Mix.csv`: M0111–M0145 plus M0087. This is the identical 36-row set identified by the additivity test in Section 2, found independently, by a method with no knowledge of the additivity result.

The mechanism is explicable and not an artifact of file corruption: it is the fingerprint of peak normalization applied *after* summing the heart and lung signals, with a measured gain range of roughly $3.5\times$ – $135\times$ across these 36 rows (not integer overflow, which was our original hypothesis before checking the gain magnitudes directly). That two independent checks — one purely numerical, one a physical artifact of the recording pipeline — converge on the same 36 IDs is, in our view, strong evidence that this is a genuine property of the released files rather than a bug in either check.

4 Sample-rate discrepancy

Every one of the 535 released files measures 4,000 Hz — mono, 16-bit, 15.0 s (60,000 frames) — without exception. The descriptor gives one sample-rate figure, and it is $5.5\times$ higher: Table I on page 138, *Technical Specification of the Digital Stethoscope*, lists “Recording Sample Rate: 22 050 Hz”.

We want to be precise about what that does and does not assert, because the distinction changes what we are asking for. Table I specifies the *instrument*, not the distributed files. The section that does describe the distributed files — *Records and Storage* — states their count, format, grouping and metadata, but never their sample rate. The only other passage that touches the rate is comparative: on page 134, immediately after noting that the earlier manikin dataset of Tsai et al. was recorded “at 8 kHz sampling rate”, the descriptor states that its own digital stethoscope

provides “improved recording quality, a higher sampling rate, and better resolution”. The released files, at 4,000 Hz, are below that 8 kHz reference, so this is the one sentence in the descriptor that the release contradicts outright. Everything else is consistent with the files having been captured at 22 050 Hz and downsampled afterwards, undocumented.

What the descriptor leaves a reader with, then, is one explicit sample-rate number and one comparative claim, and no way to learn that the files carry a rate that satisfies neither. The recording path the descriptor itself describes — stethoscope, then Bluetooth to the Eko mobile application, then that vendor’s cloud platform, then download as `.wav` — has several points at which a resampling could occur, and none of them is documented. A reader who takes 22 050 Hz as the data rate, which is the only inference the paper supports, will be wrong by a factor of 5.5.

We note explicitly that this is a property of the release and not of a redistribution, because our own first reading of it was the opposite. The copy we initially audited was obtained from github.com/Torabiy/HLS-CMDS, which left open the likelier-seeming explanation that we had measured a downsampled convenience mirror. We tested that by re-downloading the dataset from Mendeley (<https://data.mendeley.com/datasets/8972jxbmp/3>, v3, retrieved 2026-08-19) and checksumming it independently: all six top-level artifacts (HS, LS and Mix, each as `.csv` and `.zip`) are byte-identical between the two copies. The mirror is faithful; the discrepancy is in the release.

We therefore recommend the narrower fix that this finding actually supports: that *Records and Storage* state the released files’ own sample rate explicitly, and that whatever processing sits between capture at 22 050 Hz and release at 4 000 Hz be described, or the files re-issued at the capture rate if the downsampling was not intended.

The discrepancy is unlikely to have invalidated published results built on this dataset — cardiac energy below 200 Hz and respiratory energy below 1 kHz both sit comfortably under a 2 kHz Nyquist, so the retained band is adequate for the analyses these papers perform. It does affect tooling and reproduction: pretrained separation and classification models expecting 8–16 kHz input cannot be applied to these files without resampling, a reader sizing an STFT or a filter bank from 22 050 Hz will get every frequency bin wrong, and anyone reproducing a published pipeline needs to know which rate its author actually worked at.

5 Downstream impact

We are aware of four published studies that build on HLS-CMDS [2, 3, 4, 5]. We determined, from each paper’s own text, tables and figures, which `Mix.csv` rows it evaluates on (Table 1).

Two of the four therefore evaluate on the full 145 rows, and so on 109 rows whose mixed recording is not an additive combination of its listed sources. For Yaqub et al. [3] this is measurable: 4,033 of the 5,365 windows behind the 41.0% accuracy of their Table 9 carry a heart sound class label taken from a heart recording that our audit finds absent from the audio being classified. Puneet et al. [5] run their EVMD separation over the same 109 rows but report no metric against the listed sources, so the exposure there is to the illustrated separation quality rather than to a published number. The remaining two are not exposed to the additivity finding: Torabi [2] constructs its own mixtures rather than using `Mix.csv`’s, and Han et al. [4] use HLS-CMDS only to fit dictionaries. None of the four papers reports an independent integrity check of the kind described here, and two of them – Torabi’s §4.1.3 and Han et al.’s Figure 3 – do not state which recordings they used, so the exposure of those two cannot be settled from the published text at all.

We raise this not to characterize any of these results as incorrect, but because none of the four papers had the means to know whether it applied – the additivity gap is not visible from the CSV metadata or from listening to a handful of files; it required a systematic per-row check.

Table 1: `Mix.csv` rows entering each downstream study’s evaluation, as determined from that study’s own text and tables. “Outside the 36” counts rows that fail the additivity test of Section 2.

Study	<code>Mix.csv</code> rows evaluated on	Outside the 36 additive rows
Torabi [2]	None. The separation chapters build their own mixtures: §4.2.3 (p. 52) re-pairs HLS-CMDS heart and lung segments at random into 25,000 mixtures; §4.1.3 (p. 41) reports “210 clinical manikin recordings”, which is neither 145 nor identified as <code>Mix.csv</code> . §5.5 (p. 65) and §6.2.2 (p. 72) use the 50 isolated heart and 50 isolated lung recordings only.	0. The §4.1.3 evaluation set is <i>not stated</i> ; we record that as a finding rather than infer it.
Yaqub et al. [3]	All 145. <i>Not stated in the text</i> . Table 9 (p. 2528) reports 5,365 scored windows on bandpass-separated heart sounds; segmentation is into 400 ms windows (p. 2514), giving 37 whole windows per 15 s recording, and $145 \times 37 = 5365$. Its four per-class supports (481 / 2109 / 1147 / 1628) equal 37 times <code>Mix.csv</code> ’s own per-class row counts (13 / 57 / 31 / 44) exactly, for every class.	109 of 145 rows, i.e. 4,033 of the 5,365 windows (75.2%) , by class: 333 Normal, 1,628 Murmur, 851 Extra Sound, 1,221 Rhythm Disorder.
Han et al. [4]	None scored. HLS-CMDS is “used exclusively for dictionary learning and [was] not included in the respiratory disease classification experiments” (§3.2.3, p. 7). <i>Which</i> isolated recordings the dictionaries were fit on is not stated. Figure 3 (p. 8) does show one mixture beside a ground-truth respiratory signal; its provenance is not stated.	0 scored; 1 unidentified row in a qualitative figure.
Puneet et al. [5]	All 145, <i>stated</i> : the lung-isolation algorithm “is evaluated on the HLS CMS dataset,” enumerated as Normal (28), Wheezing (28), Pleural Rub (25), Rhonchi (23), Fine Crackles (22), Coarse Crackles (19) (p. 23) – summing to 145 and matching <code>Mix.csv</code> ’s lung-type distribution on every class.	109 of 145 rows (75.2%): 18 Normal, 16 Wheezing, 24 Pleural Rub, 21 Rhonchi, 13 Fine Crackles, 17 Coarse Crackles. No separation metric is reported on them; the 80.55% accuracy (p. 24) is measured on HF_Lung V1 (p. 23), not on HLS-CMDS.

6 Recommendation

We suggest the descriptor be amended, or an erratum issued, to note: (1) that only 36 of 145 `Mix.csv` rows are additive combinations of their listed sources, with row IDs enumerated above, so that downstream users can restrict evaluation to the valid subset or account for the rest — two published studies already evaluate on all 145 (Section 5); and (2) the sample-rate finding of Section 4 once resolved. We would welcome the opportunity to share our audit code directly with the authors if useful for reproducing or disputing this finding.

7 Data and code availability

The additivity check (`verify_additive_triplets()`) and the clipping-based audio-quality scan referenced above are available at https://github.com/SouthsideProgrammer/TEEP_Internship (commit 88c0ef8), and are independently re-runnable against any copy of HLS-CMDS: `make install`, `make dataset` (which verifies the six release checksums), then the audit targets. No part of the dataset is redistributed there.

References

- [1] Y. Torabi, S. Shirani, and J. P. Reilly, “Descriptor: Heart and Lung Sounds Dataset Recorded from a Clinical Manikin using Digital Stethoscope (HLS-CMDS),” *IEEE Data Descriptions*, vol. 2, pp. 133–140, 2025. doi:10.1109/IEEEDATA.2025.3566012.
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- [3] A. Yaqub, M. S. Orakzai, M. F. Qureshi, Z. Mushtaq, I. Siddique, and T. Radwan, “Spectrotemporal Deep Learning for Heart Sound Classification under Clinical Noise Conditions,” *Computer Modeling in Engineering & Sciences*, vol. 145, no. 2, pp. 2503–2529, 2025. doi:10.32604/cmescs.2025.071571.
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